2024 AMC 10A Problem 2 - Solution

Review the full statement and step-by-step solution for 2024 AMC 10A Problem 2. Great practice for AMC 10, AMC 12, AIME, and other math contests

2024 AMC 10A Problem 2

Problem:

A model used to estimate the time it will take to hike to the top of a mountain on a trail is of the form $T = aL + bG$, where a and b are constants, T is the time in minutes, L is the length of the trail in miles, and G is the altitude gain in feet. The model estimates that it will take 69 minutes to hike to the top if a trail is 1.5 miles long and ascends 800 feet, as well as if a trail is 1.2 miles long and ascends 1100 feet. How many minutes does the model estimate it will take to hike to the top if the trail is 4.2 miles long and ascends 4000 feet?

Answer Choices:

A. 240

B. 246

C. 252

D. 258

E. 264

Solution:

The given data from the first two hikes yield the system of equations

$$1.5a + 800b = 69$$
$$1.2a + 1100b = 69.$$

To solve this system, first subtract the second equation from the first equation to get $0.3a - 300b = 0$, which implies $a = 1000b$. Then the first equation becomes $1500b + 800b = 69$, from which $b = \frac{69}{2300} = 0.03$, and $a = 30$. Therefore the model is $T = 30L + 0.03G$. Substituting the values for the third hike gives $T = 30 \cdot 4.2 + 0.03 \cdot 4000 = \text{(B)} \boxed{246}$ minutes.

Note: Expressed in words, this commonly used model is "two miles per hour plus a half-hour for each 1000 feet of altitude gain."